2.6 Probability. Stochastic calculus.

1. What is the exponential distribution? What are the mean and the variance of the exponential distribution?

2. If X and Y are independent exponential random variables with mean 6 and 8, respectively, what is the probability that Y is greater than X?

3. What are the expected value and the variance of the Poisson distribution?

4. A point is chosen uniformly from the unit disk. What is the expected value of the distance between the point and the center of the disk?

5. Consider two random variables X and Y with mean 0 and variance 1, and with joint normal distribution. If $\text{cov}(X, Y) = \frac{1}{\sqrt{2}}$, what is the conditional probability $P(X > 0|Y < 0)$?

6. If X and Y are lognormal random variables, is their product XY lognormally distributed?

7. Let X be a normal random variable with mean μ and variance σ^2, and let Φ be the cumulative distribution function of the standard normal distribution. Find the expected value of $Y = \Phi(X)$.

17. If W_t is a Wiener process, find the variance of

$$X = \int_0^1 \sqrt{t} e^{\frac{W_t^2}{8}} dW_t.$$

18. If W_t is a Wiener process, what is $E\left[e^{W_t}\right]$?

19. If W_t is a Wiener process, find the variance of

$$\int_0^t s \, dW_s.$$

20. Let W_t be a Wiener process, and let

$$X_t = \int_0^t W_\tau d\tau.$$

What is the distribution of X_t? Is X_t a martingale?[2]

21. What is an Itô process?

22. What is Itô's lemma?

23. If W_t is a Wiener process, is the process $X_t = W_t^2$ a martingale?

[2] A solution to this question was given in Chapter 1 using integration by parts; a different solution will be given herein.